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MASTER 1 ECAP • SUMMARY NOTE

Knowledge, perception and acceptance of coastal risk management measures

A survey of 147 households in Le Croisic and Batz-sur-Mer (Loire-Atlantique)

Samuel Le Nenes • Supervised by Muriel Travers • June 2026

Abstract

Marine flooding and erosion pose a lasting threat to coastal households. Drawing on a survey of 147 residents of Le Croisic and Batz-sur-Mer, this study measures how their knowledge of the risk and their perception of it shape their support for protection measures. The main finding is that this effect differs from one measure to another: the best-informed households may support a measure or, on the contrary, distrust it.

Keywords: *coastal risk; risk perception; acceptance of measures; ordered logit; Loire-Atlantique.*

1. Research question and scope of the study

Storm Xynthia (2010: 47 deaths, around €2.5 billion in damage) and then Storm Ciarán (November 2023) highlighted how vulnerable the Atlantic coast is. These events cause marine flooding, shoreline retreat and sometimes cliff collapses. Climate change and sea-level rise now make this a long-term risk, even though the coast remains highly attractive: close to 12% of the French population lives there, and some municipalities see their population grow tenfold in summer. This raises the following question: how can exposed households be brought to accept and implement protection and adaptation measures? The answer does not depend only on the cost or technical design of the measures. It also depends on what households know about the risk and how they perceive it, a dimension studied in particular by Naud (2024).

"To what extent do households' knowledge and perception of coastal risk influence the adoption of protection and adaptation measures against erosion and marine flooding?"

The study area is the Guérande peninsula (Le Croisic and Batz-sur-Mer), a large part of which is classified as a red zone in the PPRL, the local coastal risk prevention plan. Based on an online survey, four composite indicators are built: objective knowledge of the risk (X1, verifiable knowledge), subjective and institutional knowledge (X2: familiarity with the PPRL, the CatNat natural disaster scheme or the Barnier Fund), perceived danger (X3) and perceived capacity to act (X4, based on the Protective Action Decision Model, PADM, of Lindell and Perry, 2004). These scores are compared with acceptance of the measures, measured on a five-point agreement scale.

2. Data

The survey collected 147 complete responses, mostly from main residents (76%) and owners (69%); 42% live less than 500 m from the shoreline. Women (75%) and respondents over 40 (82%) are over-represented: the results therefore apply to households that are probably already aware of the risk, not to the local population as a whole.

The four X scores are rescaled from 0 to 100. One gap stands out immediately: mean objective knowledge reaches 74.4/100, compared with only 38.4/100 for subjective knowledge. Households know the risk better than they think, and above all feel poorly informed about public schemes. Perceived danger (64.5/100) and perceived capacity to act (59.4/100) lie at intermediate levels.

Of the seven measures put to households, three show enough dispersion to be modelled: adapting the home (Y2: 68% in favour), insurance-based solutions (Y3: 38.8% in favour against 38.1% opposed, the only truly divisive measure) and protective structures (Y5: 77% in favour). The in-depth analysis focuses on these three variables. The overall model diagram is shown in Figure 1.

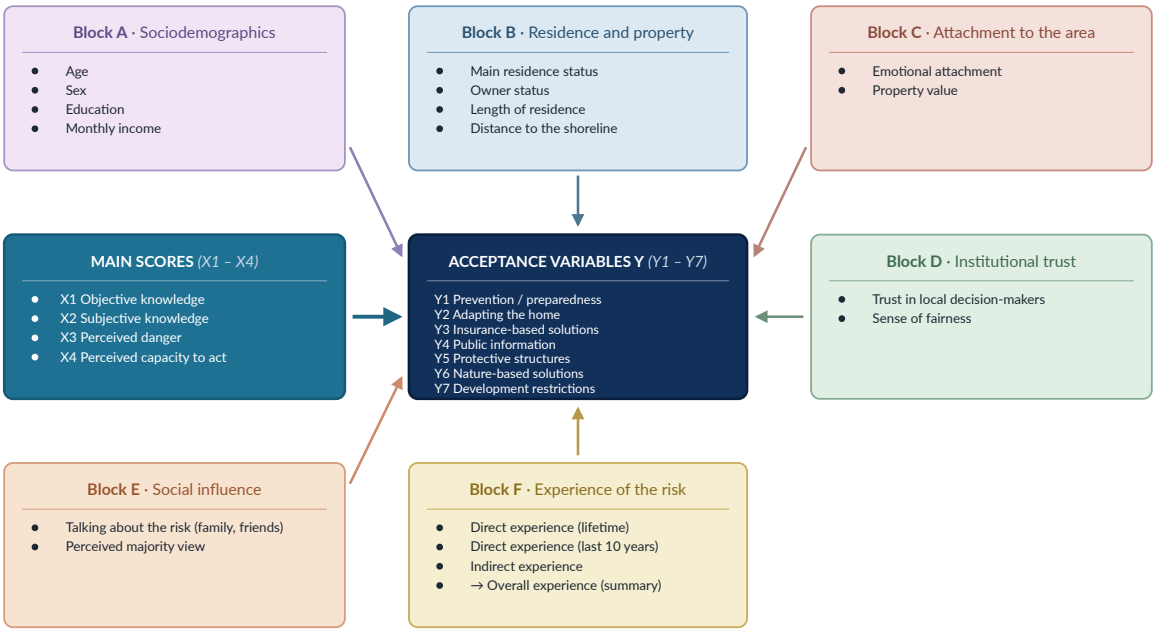


Figure 1: *overall model diagram, linking the six blocks of determinants to the seven acceptance variables (Y1 to Y7) via the four main scores (X1 to X4). Source: author's own construction.*

3. Econometric methodology

Since these three variables are ordered, the choice of method follows fairly directly. The order of the categories is known, but the gap between two successive categories is not constant, which rules out a standard linear regression. The ordered logit fits this structure: it estimates the probability that a household falls into a more favourable category from a linear combination of the explanatory variables, while respecting the order of the categories. The results are then read as odds ratios: above 1, the variable increases the odds of being more favourable; below 1, it reduces them. Finally, the model rests on the proportional odds (parallel slopes) assumption, which is checked with the Brant test.

Methodological box: *the ordered logit in one equation*

Latent variable: $Y^*_i = X_i \beta + \varepsilon_i \quad (i = 1, \ldots, n)$

Ordered thresholds: $\alpha_1 < \alpha_2 < \ldots < \alpha_{k-1}$

Cumulative probability: $\text{Prob}\,(Y_i \leq j \mid X_i) = F\,(\alpha_j - X_i \beta)$

where F is the logistic cumulative distribution function, α_j the cut-off thresholds and β the vector of coefficients, estimated by maximum likelihood. The proportional odds assumption requires β to be constant across all thresholds; if the Brant test rejects it, the constraint is relaxed locally or the variable concerned is excluded (as in the Y5_V2 model, retained after excluding objective knowledge and the majority norm).

4. Results and discussion

All three models are jointly significant, but their results differ considerably. Table 1 summarises the odds ratios of the main explanatory variables.

Explanatory variable	Y2: adapting the home	Y3: insurance-based solutions	Y5: protective structures
Cognitive dimensions (core variables)			
Objective knowledge (X1)	1.01	1.02	excluded
Subjective knowledge (X2)	0.99	1.05 ***	0.97 **
Perceived danger (X3)	1.03	1.04 **	1.02
Perceived capacity to act (X4)	1.03	0.97 *	0.99
Residential and economic situation			
Main residence	2.64 **	0.32 **	2.44 *
Owner	0.46 *	1.22	1.98
Household income	1.15	1.44 **	0.86
Distance to the sea	0.83	0.93	0.74 **
Institutional trust, norms and experience			
Trust in decision-makers	1.86 **	1.38	1.40
Perceived majority norm	2.09 *	0.84	excluded
Past experience of the risk	0.92	0.90	1.12
Overall model fit			
Pseudo-R^2 (McFadden)	0.113	0.151	0.083
Likelihood ratio test	p = 0.004	p < 0.001	p = 0.048

Table 1. *Odds ratios of the three ordered logit models. An OR above 1 increases the probability of acceptance; an OR below 1 decreases it. Significance: * p < 0.10; ** p < 0.05; *** p < 0.01. Green: significant positive effect; red: significant negative effect. "excluded": variable removed from the Y5_V2 model. Source: own estimates (R).*

Insurance-based solutions (Y3) best support the theoretical framework. This model has the best fit, with a pseudo-R^2 of 0.151 and a highly significant overall test ($p < 0.001$). Subjective knowledge has a positive effect, significant at the 1% level: ten more points on this score multiply the odds of being more favourable by about 1.57. Perceived danger works in the same direction ($p = 0.029$), as does income ($p = 0.026$), which makes sense since insurance involves a cost and coverage in proportion to the value of one's property. The result for main residence is more unexpected: it has a negative effect (OR = 0.32; $p = 0.017$). One interpretation is that main residents probably see insurance as an additional burden on an already tight budget.

For adapting the home (Y2), the results are different. Here, none of the four cognitive scores is significant, so the hypothesis of a direct effect of knowledge or capacity to act is not supported. Other variables drive the result: main residence status (OR = 2.64; $p = 0.033$), trust in local decision-makers (OR = 1.86; $p = 0.025$) and, marginally, the perceived majority norm (OR = 2.09; $p = 0.051$). Agreeing to invest in one's home therefore depends mainly on the household's actual situation and on trust, more than on how well informed it is.

Protective structures (Y5) produce yet another pattern, and the most surprising one. Only three variables stand out: distance to the sea (OR = 0.74; $p = 0.016$), meaning that the closer people live to the shore, the more they accept structures; main residence (OR = 2.44; $p = 0.056$); and above all subjective knowledge, this time with a negative sign (OR = 0.97; $p = 0.013$). This is the opposite of insurance: the best-informed households are more critical of structures whose protection is visible but imperfect, and which can create a false sense of security.

5. Conclusion, assessment and recommendations

Three lessons emerge. **(1)** Knowledge does not have a uniform effect: the same variable (subjective knowledge) increases acceptance of insurance but reduces acceptance of protective structures. “Better information” is therefore not enough to win acceptance. **(2)** It is subjective knowledge that matters, not objective knowledge (never significant): how far people have taken the schemes on board counts for more than verifiable knowledge. **(3)** Residential, institutional and social variables carry as much weight as the cognitive scores, if not more. Main residence in particular is the only variable that is significant in all three models, but with opposite signs depending on the measure.

Recommendations for policymakers. Approaches need to differ from one measure to another. For insurance, targeted communication on the schemes, such as CatNat, the Barnier Fund and eligibility conditions, would close the gap between objective and subjective knowledge. For adapting the home, the focus should rather be on trust and practical support, for example participatory schemes run by coastal inter-municipal authorities (EPCI), such as CEREMA’s *ateliers des territoires*, and clear financial aid, which seem more effective than top-down information. Finally, for protective structures, transparent communication about their limits, in particular maintenance costs and declining effectiveness as sea levels rise, would better support lasting acceptance than a promise of absolute safety.

Limitations. The sample is small (147 households) and made up of respondents who are probably already aware of the risk, and the models identify associations, not causal relationships. A survey repeated over time, combined with a qualitative component, would strengthen these conclusions.

Theoretical scope. These results suggest broadening the initial framework, which focuses on cognitive dimensions (PADM model): institutional trust, perceived social norms and residential status are not just control variables but important determinants of acceptability, sometimes more so than the cognitive scores themselves. Including them explicitly would better account for the different reasons for support observed from one measure to another, and help design public policies suited to each type of scheme.

References

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Key takeaways: three acceptance patterns, three policy levers

Y3	**Insurance-based solutions: the cognitive dimensions work.** Subjective knowledge and perceived danger increase acceptance. **Lever:** communicate on the CatNat scheme, the Barnier Fund and eligibility conditions.
Y2	**Adapting the home: contextual factors come first.** Main residence, trust and social norms dominate; no cognitive score is significant. **Lever:** local participatory schemes and clear financial support.
Y5	**Protective structures: the logic is reversed.** The best-informed households are more critical of them (negative effect of X2). **Lever:** transparency about their limits rather than a promise of absolute safety.